Topic 1 Review · IEP

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · Tuesday, Nov 17 · Test Wed Nov 18

Name _____ Period _____ Date _____

Directions: This review has the same kinds of questions as the Topic 1 test, in the same order. Use the tips and fill in each step. Choose **one** answer for multiple choice. Finish Part 1 (just like the test) first; Part 2 is extra practice. One question per page. You may use the reference box and a calculator. Ask if you need a question read aloud.

Reference box

Solving: distribute → combine like terms → get the variable on one side → undo + and − → undo × and ÷. Check by substituting.

Same sides ($3x + 5 = 3x + 5$): infinite solutions · **same x-term, different numbers** ($3x + 5 = 3x + 2$): no solution.

Inequalities: multiply or divide by a negative → **flip** the sign. < or >: open circle · ≤ or ≥: closed circle.

Slope-intercept form: $y = mx + b$ · Standard form: $Ax + By = C$ · Triangle area = ½bh · Rectangle area = ℓw

Part 1 · Just like the test (Questions 1–11)

1. Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, y, after x weeks?

> **Tip:** Rate × weeks + starting amount.

Step 1: Rate per week = _____

Step 2: Starting amount = _____

Step 3: y = _____x + _____

A. $y = 15 + 40x$

B. $y = 15x + 40$

C. $y = -15x + 40$

2. What value of k makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

> **Tip:** Distribute, then combine like terms on each side. Same variable terms + different numbers = no solution.

Step 1: Left side: $5 - 4k - 12 =$ _____

Step 2: Right side: $9 - 4k - 12 =$ _____

Step 3: $-7 = -3$ is _____, so: _____

F. 1

G. 2

H. No solution

3. What is the solution set for $3(4 - t) > -2t + 15$?

> **Tip:** Dividing or multiplying by a negative **flips** the inequality sign.

Step 1: Distribute: _____ $> -2t + 15$

Step 2: Add $2t$, subtract 12: _____ > _____

Step 3: Divide by -1 and flip: t _____ _____

A. $t > -3$

B. $t > 3$

C. $t < -3$

4. Solve the equation shown.

$0.3(4x - 6) = -1.8x + 4.2$

> **Tip:** Distribute the 0.3 to both terms first.

Step 1: Distribute: _____ − _____ = $-1.8x + 4.2$

Step 2: Add $1.8x$ and 1.8: _____x = _____

Step 3: x = _____

Answer: _____

5. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

> **Tip:** Simplify each side. Identical sides = infinite. Same x-term, different numbers = none.

Step 1: Left: $\frac{1}{3}(12x) + \frac{1}{3}(30) =$ _____

Step 2: Right: $4x + 8 + 2 =$ _____

Step 3: Answer: _____

Answer: _____

6. Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, y, when Maya buys x pens.

> **Tip:** The thing that changes (pens) gets the x. Fixed items become the constant.

Step 1: Notebooks: 4 × 1.75 = _____

Step 2: Pens: _____x

Step 3: y = _____x + _____

Answer: _____

7. Rewrite the equation in slope-intercept form.

$2x + 5y = 15$

> **Tip:** Get y alone: move the x-term, then divide **every** term by the number in front of y.

Step 1: Subtract $2x$: $5y =$ _____ + 15

Step 2: Divide every term by _____

Step 3: $y =$ _____$x +$ _____

Answer: _____

8. The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for h.

> **Tip:** Undo what is being done to h. It is multiplied by ℓw, so divide.

Step 1: h is multiplied by _____

Step 2: Divide both sides by _____

Step 3: h = _____

Answer: _____

9. Solve the inequality, and graph the solution on the number line.

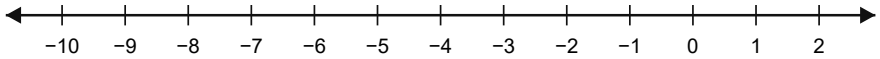
$2.5m - 3 < -1.5(m + 18)$

> **Tip:** Open circle for < or >. Closed circle for ≤ or ≥.

Step 1: Distribute: $2.5m - 3 <$ _____

Step 2: Add $1.5m$ and 3: _____ $m <$ _____

Step 3: m _____ _____ → _____ circle at −6, shade _____



Answer: _____

10. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

> **Tip:** Distribute, then add the number on the y side to both sides.

Step 1: Distribute: $y - 4 =$ _____

Step 2: Add _____ to both sides

Step 3: $y =$ _____

Answer: _____

11. A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of n?

> **Tip:** Triangle area = ½ · base · height. Rectangle area = length · width. Set them equal.

Step 1: Triangle: $\frac{1}{2} \cdot 10 \cdot (3n - 6) =$ _____

Step 2: Rectangle: $5(n + 4) =$ _____

Step 3: $15n - 30 = 5n + 20 \rightarrow n =$ _____

Answer: _____

Part 2 · Extra practice (Questions 12–15)

12. Which equation has **no solution**?

> **Tip:** Simplify each one. Look for the same x-term but different numbers.

F. $3(x - 2) = 3x + 5$

G. $3(x - 2) = 3x - 6$

H. $3(x - 2) = 2x + 5$

13. What is the solution set for $-4x + 7 \geq 19$?

> **Tip:** Divide by a negative → flip the sign.

Step 1: Subtract 7: $-4x \geq$ _____

Step 2: Divide by −4 and flip: x _____ _____

A. $x \geq -3$

B. $x \leq 3$

C. $x \leq -3$

14. The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for w.

> **Tip:** Undo the adding first, then the multiplying.

Step 1: Subtract 2ℓ: $2w =$ _____

Step 2: Divide by _____: $w =$ _____

Answer: _____

15. A gym charges a $25 sign-up fee plus $8 per class. Lena can spend at most $90. Write and solve an inequality for the number of classes, c, she can take.

> **Tip:** “At most” means ≤.

Step 1: $25 + 8c$ _____ 90

Step 2: $8c \leq$ _____

Step 3: $c \leq$ _____ → at most _____ classes

Answer: _____